SWEEP rule on the KD-Toric code

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Move this definition to the beginning of the paper.

Definition 0.1. We model the qubits on the finite toric with a side length n by the projection:

$$(x_1, x_2, .., x_d) \mapsto (x_1 \bmod n, x_2 \bmod n, .., x_d \bmod n)$$

Rewrite when we talk about the first time we have deviation clusters of size $\in (\alpha d, d)$. The reason is that having those assumptions means all the droplets in the past shrank; thus, we can use the ratio of edges to time-leaves in the big subgraph.

Claim 0.2. Assume that until step τ, there was no a deviation cluster of size greater than d. Let $\hat{K}$ be a spanned slice induced by a deviation of a cluster state at step τ and let ω be an explanation for $\hat{K}$. Assume that:

1. The number of edges in w is greater than βn.
2. For any spanned slice $\hat{K}' \in \omega$ such $\hat{K}' \neq \hat{K}$ it holds that $\mathbf{Size}(\hat{K}') < \delta n$.

Then there is a subgraph (subtree) ω' of ω such that ω' has less than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ leaves.

Claim 0.3. Consider a probability space, $(\Omega, \mathbf{Pr})$, for which the atomic events are subgraphs of $G = (V = \mathbb{Z}^d \times \mathbb{Z}_t, E)$, where two vertices might be connected only if their difference in the time coordinate is at most 1 and their Hamming distance in the space coordinate is at most 2. We say that a vertex of such a subgraph is a time-leaf if and only if it is not connected to vertices with a lower time coordinate. Notice that an outcome $\omega \in \Omega$ does not have to be connected. Suppose that the following three conditions hold:

1. There exists $n \in \mathbb{N}$ such that ω is invariant under shifting by $n \cdot \mathbf{1}_i$ for any $i \in [d]$.
2. The probability function $\mathbf{Pr}$ induces a locally stochastic noise bound over the time-leaves, which are different in their space coordination module n. Namely, there is $p \in (0, 1)$ such that for any $S \subset V$, with a maximal subset $S' \subset S$, such that for any $v = (v_x, v_t), u = (u_x, u_t) \in S'$ it holds that $v_x \mod n \neq u_x \mod n$, the probability that S are time-leaves of an outcome subgraph is less than:

$$\mathbf{Pr}\left[S \text{ are time-leaves } \right] \leq p^{|S'|}$$

3. For any subgraph $\omega \in \Omega$, and for any $\omega' \subset \omega$ such ω' is connected. The ratio of edges to time-leaves of ω' is at most γ.

Then there is $p_c \in (0,1)$ such that for $p < p_c$, the probability of sampling a subgraph with more than βn vertices is less than:

$$tn^d q^n$$

Proof. Let $\omega \in \Omega$ with more than βn vertices, observes that a spanning tree of ω has a ratio of edges to time-leaves which is at most ω's ratio, and number of edges equals to ω's vertices - 1. In addition the time-leaves of such spanning tree are just leaves. Therefore, by **??**, ω has a subtree with less than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ time-leaves. Now, since any two connected vertices are at a space-distance of at most 1, it follows that all the leaves in the subtree are at a space-distance of at most βn from each other. Thus, if $\beta < 1$, no two of them have the same space-coordinate modulo n. And therefore the probability to have such subtree is at most $p^{\frac{\beta}{4\gamma}n}$.

Because any ω with more than βn vertices implies the existence of a subtree with size less than $\frac{\beta}{2}n$, it's enough to bound the probability of having such a subtree. Considering that the trees are invariant to shifting by $n \cdot \mathbf{1}_i$, we have that it's sufficient to show that there are no such trees which are rooted in the finite cube $\mathbb{Z}_n^d \times \mathbb{Z}_t$. Finally, we use **??** to bound the number of such trees given a fixed root. Overall, we get:

$$\mathbf{Pr}\left[\text{No } \omega\text{s with more than } \beta n \text{ vertices}\right] \leq n^d t \left(\xi^2 p^{\frac{\beta}{4\gamma}}\right)^n$$

$\square$